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STA301 QUIZ#4



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1. The standard deviation of sampling distribution of proportion is:
   a. n/p
   b. p
   c. q/n
   d. √(pq/n)
2. As a rule of thumb, when n>=30, then we can assume that.....is normally distributed:
   a. Sampling distribution of sample mean
   b. Probability distribution
   c. Binomial distribution
   d. Sampling distribution
3. **Which of the following is most important and most widely used method in point estimation?**
   a. The method of fractional moments
   b. The method of maximum likelihood
   c. The method of least square
   d. The method of moments
4. **In sampling from a large population with б = 35, the standard error of the mean is found to be 4. The size of the sample used is:**
   a. 77 **(HB)**
   b. 15
   c. 56
   d. 25

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5. As the sample size _____ the variation of the sampling distribution of the same means _____:
   a. Decreases, decreases
   b. Decreases, remains the same
   c. Increases, decreases
   d. Increases, remains the same
6. Method of Moments for estimation of parameters is based on the theory of
   a. None of the above
   b. Non-linear estimation
   c. Both Linear and Non-linear
   d. Linear Estimation
7. A confidence interval has a specified probability_____ of containing the true value of the parameter.
   a. 1-α (HB)
   b. 1+α
   c. α/2
   d. α

HB
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8. The mean of the sampling distribution of x¯1−x¯2, denoted by μx¯1−x¯2is equal to the difference between repectiveThe mean of the sampling distribution of $\bar{x}_1-\bar{x}_2$, denoted by $\mu_{\bar{x}_1-\bar{x}_2}$is equal to the difference between repective

9.

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a. Sample Proportions
b. Population Proportions
c. Population Means
d. Sample Means (HB)

10. If Var(T1) >Var(T2) ,where T1 and T2 are two unbiased estimators ,then:
a. T2 is more efficient
b. T1 is more consistent
c. T2 is more consistent
d. T1 is more efficient

11. If our sampled population is normal, then sampling distribution of the sample mean will .....:
a. None of the above
b. Normal
c. Depends on sample size
d. Not necessarily normal

12. When sample size is to be considered large, population standard deviation can be replaced by
a. Sample Mean
b. Sample Variance
c. Sample Standard Deviation (HB)
d. Population variance

13. The standard error of the sampling distribution of p1 - p2 is equal to:
a. None of the above

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b. √(p/n)
c. √(p1q1/n1)+ p2q2/n2
d. √((p1+p2)/n)

14. In general, the estimators obtained by the method of moments are .....
a. All the above
b. Biased
c. Inconsistent
d. Inefficient

15. As sample size goes up, what tends to happen to 95% confidence intervals?
a. They become more precise and narrow
b. They become more precise
c. They become more narrow
d. They become wider

16. P 0 1 2 3
f(p) 1/20 9/20 9/20 1/20
The mean of the sampling proportion will be:
a. 25/20
b. 10/20
c. 15/20
d. 30/20 (HB)

17. Consistency is a ..... property.
a. Non-random sample
b. Large sample
c. Unique sample
d. Small sample

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18. An estimator is said to be..... if its expected value is equal to true value of its parameter.
    a. Unbiased (HB)
    b. Consistant
    c. Efficient
    d. Biased
19. Method of Moments for estimation of parameters is based on the theory of
    a. Non-linear estimation
    b. Linear Estimation (HB)
    c. Both Linear and Non-linear
    d. None of the above
20. In sampling without replacement; fpc is used when:
    a. None of the above
    b. n=0.05N
    c. n<0.05N
    d. n>0.05N
21. Assume a large distribution in which б = 6. What is the standard error of the mean based on samples for which n = l0?
    a. l67
    b. 1.9 (HB)
    c.15
    d.3

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22. The difference between the value of the statistic and the value of the parameter is called _____.

a. Sampling Error
b. Biased Error
c. Random Error
d. Standard Error

23. When an interval is constructed, what is the probability of occurrence of parameter in it? (HB)

a. Either 1 or 0
b. 50
c 1.
d. 0

If the sampled population has a normal distribution, when is the sampling distribution of the sample mean also a normal distribution?

c. Only when the sample size is smaller than 30
d. Only when the sample size is at least 30
e. Only when the sample size is at least 100
f. Always

24. Method of Moments for estimation of parameters is based on the theory of (HB)

a. Linear Estimation
b. Non-linear estimation
c. Both Linear and Non-linear
d. None of the above

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25. An industrial designer wants to determine the average amount of time it takes an adult to assemble an "easy to assemble" toy. A sample of 16 times yielded an average time of 19.92 minutes, with a population standard deviation equal to 5.73. Assuming normality of assembly times, what is a 95% confidence interval for the mean assembly time?

a. (17.1123, 30.7277)
b. (17.1123, 24.7277)
c. (14.1123, 22.7277)
d. (17.1123, 22.7277) (HB)

26. Which of the following is the most common example of a situation for which the main parameter of interest is a population proportion?

a. A randomized experiment
b. An observational study
c. A binomial experiment (HB)
d. A normal experimen

27. "To consider every possible value that the parameter might have, and for each value, compute the probability and THAT value of the parameter for which the probability of a given sample is greatest, is chosen as an estimate." This procedure is known as (HB)

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a. The method of maximum likelihood
b. The method of least square
c. The method of fractional moments
d. The method of moments

28. A consistent estimator will always be a:

a. May or may not be unbiased estimator
b. Unbiased estimator
c. Biased estimator
d. Unbiased estimator and biased estimator

29. The standard deviation of sampling distribution of proportion is:

a. p
b. n/p
c. q/n
d. √(pq/n) **(HB)**

30. If an estimator is more efficient then the other estimator, its shape of the sampling distribution will be

a. Highly peaked
b. Skewed to right
c. Flattered
d. Normal

31. If the NULL hypothesis says that the average height of Pakistani soldiers exceeds the average height of American soldiers by not more than 3 inches. What will be its ALTERNATIVE hypothesis?

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a. P > 3
b. P < 3 (HB)
c. none of these
d. P = 3

32. Mean of t-distribution will not exist when d.f(degree of freedom) will equal to:
a. 1 (HB)
b. 2
c. 3
d. 4

33. The location of the critical region depends upon:
a. Value of alpha (HB)
b. Null hypothesis
c. Value of test-statistic
d. Alternative hypothesis

34. In testing H0: μ = 100 against H1: μ ≤ 100 at the 10% level of significance,H0 is rejected if:
a. The p-value is less than 0.10
b. 100 is contained in the 90% confidence interval
c. The p-value is greater than 0.10
d. The value of the test statistic is in the acceptance region

35. The proportion of males in Pakistan is at least 0.48, the Null Hypothesis is p ≥ 0.48, then the alternative hypothesis H1 is

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a. $P \geq 0.48$
b. **P < 0.48 (HB)**
c. **P = 0.48**
d. $P \leq 0.48$

36. If we are testing H0 : mean = 50 against H1 : mean < 50 then the rejection region will be
a. on both sides
b. in the center
c. on right side of mean
d. **on left side of mean**

37. The proportion of working females in Pakistan is at most 0.40, the Null Hypothesis is H0 : P≤ 0.40. then the alternative hypothesis H1 is
a. **P > 0.40 (HB)**
b. P < 0.40
c. P = 0.40
d. none of these

38. We can apply t-distribution to test the hypothesis, when
a. Sample is small
b. **Population variance is unknown and Sample is small**
c. Population variance is known
d. Sample is large

39. The degrees of freedom associated with paired t-test .....
a. n1-2 and n2-2

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b. n-2
c. n1+n2-4
d. n-1

40. For a particular hypothesis test, α=0.05and β=0.05. The power of test is equal to: (HB)
a. 0.90
b. 0.05
c. 0.95
d. 0.50



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